

Same Death Star. Different Blueprints.



KATY IN CONTROL — EPISODE 2 COMPANION GUIDE



"Same mission. Same three layers. Two very different ways of running the Death Star."

PREVIOUSLY ON...

Previously on Katy In Control...

You learned the three layers that run every great control system in the galaxy. Here's your lightning recap before we go deeper.

The Droids

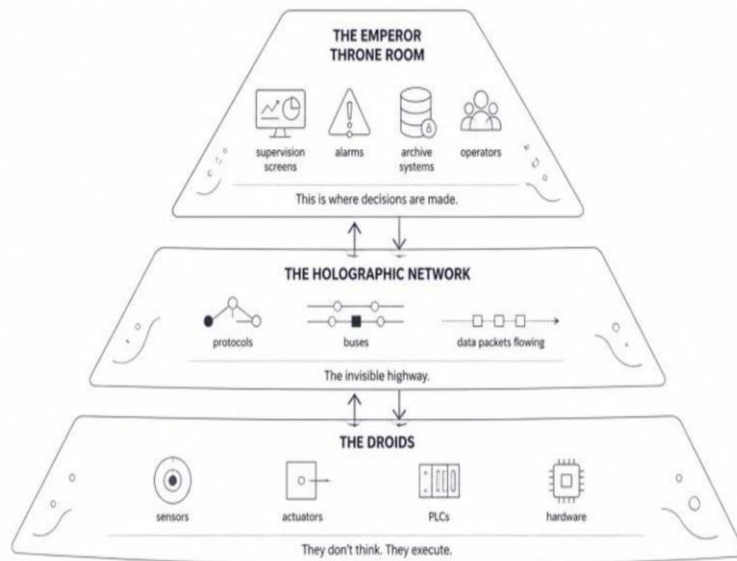
Sensors and actuators. They observe — or they act. They don't think. They don't question orders. They just **execute**.

The Holographic Network

The communication bus. Data goes **up**. Orders go **down**. It never sleeps. It never stops. The Empire runs on this.

The Emperor's Throne Room

The SCADA server. Sees everything. Decides everything. **Executes immediately**. No second-guessing.



You know the three layers. Now meet the two galaxies. 

Same Data Point. Two Very Different Philosophies.

● EPICS SIDE

PROCESS VARIABLE

Short: PV

Name it whatever you want. No rules. No database. No referee. Two PVs can share the same name on the same network. The team decides the conventions. Or doesn't.

```
temperature
my_temperature
TEST_temp_v2
// same network. no conflict detected.
```

⚡ FLEXIBLE — POWERFUL, BUT WATCH YOUR NAMING.

● TANGO SIDE

ATTRIBUTE

Every value belongs to a Device. Every Device has a unique address. The database enforces uniqueness. No duplicates. Ever. By design.

DOMAIN / FAMILY / DEVICE

Attribute: temperature

LHC / Cryogenics / Valve01

temperature // unique. guaranteed.

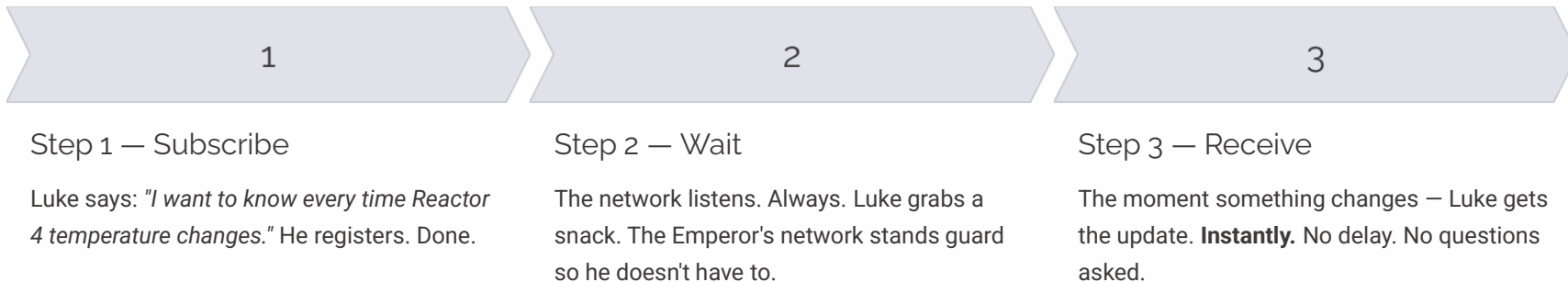
✅ STRUCTURED — UNIQUENESS ENFORCED BY CONTRACT.

"In EPICS, naming is a convention. In TANGO, it's a contract. Two philosophies. Both valid. We'll go deeper in a future episode."

EPICS — THE NETWORK LAYER

How EPICS Sends a Message

No polling. No asking. No "hey, are you there?" Just pure, elegant **listening**. EPICS uses a publish/subscribe model called **Channel Access** (CA) — upgraded in modern deployments to **PV Access**.



📄 ⚡ **Channel Access** → **PV Access**. Same idea. Full HD transmission. Backward compatible. The Empire upgraded its hologram codec.

TANGO — THE NETWORK LAYER

How TANGO Sends a Message

TANGO didn't start with one protocol. It started with **CORBA** — structured, reliable, very Imperial — and evolved into something much more Force-sensitive: **ZeroMQ**.

CORBA — The Classic

"Send a messenger droid. Wait for it to come back. Ask: 'Hey, what's the current value?'"

Reliable. Structured. Very Imperial. A little slow. Works great — until the galaxy gets impatient.

POLLING MODEL

ZMQ — The Modern

"Don't ask. Just feel the change — the moment it happens. Like a direct Force connection."

Event-driven. Blazing fast. TANGO's glow-up moment. From messenger droid to Force lightning.

EVENT-DRIVEN MODEL

TANGO started with CORBA. Evolved to ZMQ. From messenger droid to direct Force connection. 

EPICS — THE BRAIN LAYER

Darth Vader's Command Ship — The IOC

Meet the **IOC** — Input Output Controller. A software program running on a server. It commands every Process Variable under its watch. It exposes them to the network. Every. Single. One.



READ Door

Temperature too high? **READ**. The IOC reports it to the network. Accurate. Instant. Unwavering.



WRITE Door

Change a setpoint? **WRITE**. The IOC accepts the command. Applies it. Confirms it. The Empire obeys.

A PV enters through **one door only** — READ or WRITE. Not both simultaneously. That's not a bug. That's architecture by design.



Battle-tested in particle accelerators, fusion reactors, space telescopes. The IOC doesn't complain. It just works.

TANGO — THE BRAIN LAYER

The Device Server — and the Secret Weapon

The **Device Server** is TANGO's IOC equivalent. Attributes in read. Attributes in write. Looks familiar so far, right? But then — TANGO pulls out something EPICS doesn't have natively.



Read Attribute

Same as EPICS. The Device Server exposes a value. The network reads it. Clean and simple.



Write Attribute

Same as EPICS. Send a new value. The Device Server accepts it. Business as usual.



COMMAND

This is the secret weapon. One call. One action — or an entire chain of events across the galaxy. Like Order 66, but hopefully for science.

"A Command can be simple: 'Turn on.' One call. One action. Or a Command can be Order 66. One call. The galaxy changes in seconds."

That's the TANGO Command. One call. Potentially a LOT of consequences. Choose wisely. 

THE COMPARISON

Same Death Star. Two Blueprints.

Mission briefing time. Everything you need. One table. Dark background. No excuses for confusion.

CONCEPT	 EPICS	 TANGO
Data point name	Process Variable (PV)	Attribute
Classic network	Channel Access	CORBA
Modern network	PV Access	ZeroMQ / ZMQ
Brain unit	IOC	Device Server
Extra power	—	 Commands
Network style	Subscribe & receive	Poll → then push
Philosophy	Simple. Scalable. Global.	Object-oriented. Elegant. European.

Different words. Same problems. Both remarkable. Both battle-tested. Both running real scientific facilities right now. 

KEY TAKEAWAYS

What You Now Know, Padawan. 🎓

⚡ Same Droid, Different Badge

EPICS calls it **PV**. TANGO calls it **Attribute**. Same Droid under the hood — different name tag pinned to its chest.

🔬 Two Network Philosophies

EPICS uses **Channel Access** — subscribe and receive. TANGO started with **CORBA**, evolved to **ZMQ**. Messenger droid → Force connection.

🧠 IOC vs. Device Server

EPICS runs **IOCs** — PVs in read OR write. TANGO runs **Device Servers** — plus the secret weapon you're about to memorize forever.

🔴 Commands Are TANGO's Superpower

Simple action — or Order 66. One call. Your choice. EPICS doesn't have this natively. That's not a flaw — that's a design philosophy.

👥 Both Galaxies Are Remarkable

Two communities. Two continents. One shared goal: run the most advanced scientific facilities on the planet. The galaxy is big enough for both.

EPISODE 3 — COMING UP NEXT

Every Great System Has an Origin Story.

● EPICS — The American Story

Los Alamos National Laboratory. Late 1980s. Bob Dalesio and a small team decide they're tired of reinventing the wheel — and build EPICS instead. The American way: make it scalable, make it open, make it work.

● TANGO — The European Story

ESRF Grenoble. Late 1990s. Andy Gotz and collaborators look at the landscape and think: "*We can do this — but elegantly.*" Object-oriented. Structured. Very French. Very good.

In **Episode 3**, we travel back in time. Who built EPICS? Who built TANGO? And why did two teams, on two continents, decide to solve the same problem — in their own way? That story is coming up next.

May the uptime be with you.



Katy in Control — Real-World Industrial Systems Explained.

katysaintin.github.io